

AI technology, modern electronics warfare is becoming more intelligent, effective, and precise in many ways. The recent killing of Iranian major general Qasem Soleimani near Baghdad International Airport by a US drone in January this year was an example of the preciseness and effectiveness of a drone and also an eye-opener. According to some news, the world's most feared drone, MQ-9 Reaper, had killed Soleimani. The recent tension between the US and Iran has witnessed the use of many deadly drones in the Middle East.

Every country has its own secret planning and strategy to modernise warfare equipment in the airspace, ground, and sea. Leveraging AI technology in warfare equipment gives greater advantage in the race towards modernisation of military equipment. All the autonomous machines have some form of AI and other sophisticated technologies involved. This is why an autonomous drone is one of the most advanced and secret projects of the defence department in many countries.

This article covers the use of AI in military applications, including skies, lands and seas, and on-going modernisation of Indian warfare equipment.

AI in sky

Skyborg drone, originally designed as a flying AI machine, is now envisioned as a combat aerial vehicle having the capability to carry out dangerous missions, including hunting down or jamming enemy air defence networks, reconnaissance missions, or striking targets in heavily defended airspace. Earlier, Skyborg was meant to operate either together with a piloted fighter, providing assistance to a human pilot, or as an AI flying machine on autonomous mode. AI-powered Skyborg drones are likely to be put into use by 2023.

XQ-58 Valkyrie drone can fly at a high subsonic speed and take off without a runway. It is a stealthy drone designed



MQ-9 Reaper drone (Credit: en.wikipedia.org)

and built by Kratos Defence & Security Solutions for the US Air Force. It has been designed to perform and operate as a fighter jet. Further, the air force is exploring to use F-35 and F-15EX fighter jets together with the Valkyrie drone.

MQ-9 Reaper, developed by General Atomics Aeronautical Systems, is the first hunter-killer UAV designed for long-endurance and high-altitude surveillance. It is also called Predator B, capable of being remotely controlled or autonomous flight operations, primarily for the US Air Force. It has a range of 1,850km and an operational altitude of 15km, which makes it useful for long-term loitering operations for both surveillance and support of ground troops. The Reaper has been

sold to many countries, including the UK, France, and Australia.

AI on ground

Some countries are replacing foot soldiers on the ground with intelligent robots, allowing them to engage with enemies without putting human lives at risk. Robot tanks, vehicles, and sentries are being developed to replace human soldiers.

For example, the Uran-9 robot is an unmanned combat ground vehicle (UCGV) developed and produced by the Russian Defence company, UPTK. The robot is especially designed to deliver combined combat, reconnaissance, and counter-terrorism units with remote reconnaissance and fire support. It can be used autonomously on predefined paths or manually operated on roads.

The robotic system is equipped with a wide range of remote-controlled sensor modules such as laser warning systems and electro-optic and thermal imaging cameras. The onboard fire control system consists of automatic target detection, identification, and tracking devices as well as ballistic computers. It is capable of detecting and tracking enemy targets at a distance up to 6km during day and 3km at night. The Uran-9



Uran-9 robot (Credit: armyrecognition.com)